

**This practice is built exactly like the real test – same questions, same order, different numbers. Work every problem, then check the worked key. Anything you miss is exactly what to study.**

1. Write the equation, in slope-intercept form, of the line with a y-intercept of  $-4$  and a slope of  $5$ .
2. Write the equation, in slope-intercept form, of the line through  $(3, 4)$  and  $(5, 14)$ .
3. Write the equation, in point-slope form, of the line through  $(-4, -9)$ , with a slope of  $5/2$ .
4. Write the equation, in point-slope form, of the line through  $(5, 4)$  and  $(9, 16)$ .
5. Rewrite  $y + 5 = 4(x - 3)$  in slope-intercept form.
6. Rewrite  $y = -(5/6)x + 4$  in standard form (integer coefficients).
7. Rewrite  $14x - 7y = 21$  in slope-intercept form.

## Algebra 1 – Module 5 Practice Test

Full Name: \_\_\_\_\_

Period: \_\_\_\_\_

- 8.** Write the equation, in slope-intercept form, of the line that passes through  $(-2, 5)$  and is PARALLEL to  $y = -4x + 1$ .
- 9.** Write the equation, in slope-intercept form, of the line that passes through  $(3, -2)$  and is PERPENDICULAR to  $y = (3/4)x + 6$ .
- 10.** Classify each pair of lines as parallel, perpendicular, or neither. Justify with slopes.
- a.**  $y = -(3/7)x + 5$  and  $y = -(3/7)x - 2$
- b.**  $y = (1/6)x - 4$  and  $y = -6x + 1$
- 11.** Error Analysis — Rex says a line perpendicular to  $y = -(3/8)x + 5$  has slope  $-8/3$ . Identify his mistake and give the correct slope.
- 12.** Marisol rents a kayak for an \$8 launch fee plus \$7 per hour. Write an equation for the total cost  $C$  after  $h$  hours, then find the cost after 5 hours.
- 13.** Write the equation, in STANDARD form, of the line with x-intercept  $(5, 0)$  and y-intercept  $(0, -4)$ .

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## Cumulative Review — ACT Style. Circle the best answer.

- 14.** What is the slope of the line through  $(1, -3)$  and  $(3, 11)$ ?
- Ⓐ  $1/7$   
Ⓑ 7  
Ⓒ 14  
Ⓓ 28
- 15.** Let  $f(x) = 6x - 5$ . What is  $f(4)$ ?
- Ⓔ 14  
Ⓕ 19  
Ⓖ 24  
Ⓗ 29
- 16.** Solve:  $3(x - 4) + 2x = 28$
- Ⓐ 4  
Ⓑ  $32/5$   
Ⓒ 8  
Ⓓ 40